

Scenario 1:

Flight Delay Analysis

An airline tracks flight delays (in minutes) for 20 flights. Analyze the flight delays to calculate percentiles, detect outliers, and evaluate the overall distribution.

##Answer:

```
import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

###percentile & Outliers calculation:

Q1=np.percentile(data, 25)

Q3=np.percentile(data, 75)

IQR=Q3-Q1

Outliers=[x for x in data if (x<(Q1-1.5*IQR) or x>(Q3+1.5*IQR) ) ]

##To see the overall distribution

sns.histplot(data, kde=True)

plt.show()
```

Scenario 2:

Employee Salary Analysis

A company wants to analyze the salary distribution of its employees to understand the central tendency and determine whether the data is skewed.

##Answer:

To understand central tendency and determine whether the data is skewed.

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
sns.histplot(data, kde=True)
```

```
plt.show()
```

If the peak is on the left side of the chart, and the "tail" stretches out long toward the **right** (higher values) then it is **Right (Positive) Skew**, Which means $\text{Mean} > \text{Median}$

If the peak is on the right side of the chart, and the "tail" stretches out long toward the **left** (lower values) then it is **Left (Negative) Skew**, Which means $\text{Mean} < \text{Median}$

If the data looks like a symmetrical bell curve where both tails look identical then it is **No Skew**, which means $\text{Mean} = \text{Median}$ (approximately)

Scenario 3:

Product Sales Analysis

A retail store records product sales over 15 days. Create a frequency distribution table and visualize the sales data using appropriate charts.

##Answer:

```
import pandas as pd
```

```
df = pd.DataFrame(data, columns=['Sales'])
```

```
# Base frequency
```

```
freq_df = df['Sales'].value_counts().reset_index() freq_df.columns = ['Sales',  
'Frequency']
```

```
# Relative Frequency (Proportion)
```

```
freq_df['Relative Frequency'] = df['Sales'].value_counts(normalize=True).values
```

```
# Cumulative Frequency
```

```
freq_df['Cumulative Frequency'] = freq_df['Frequency'].cumsum() print(freq_df)
```

Scenario 4:

Student Exam Performance Analysis

A school wants to analyze the exam performance of students across three subjects: Mathematics, Science, and English. How can Data Science concepts be applied to understand their performance?

##Answer:

Using Central tendency, calculate mean to know the average in each subject. Based on the average, the teaching should be shifted.

By calculating STD, can understand the gap between toppers and failed students. Based on the gap, the special teaching care needs to be taken to bridge the gap.

By bar chart, can compare pass and fail across all 3 subjects simultaneously

Scenario 5:

Clinical Trial for Diabetes Medication

A pharmaceutical company conducted a clinical trial with two groups: one receiving medication and the other a placebo. Perform a hypothesis test to determine the effectiveness of the medication.

##Answer:

Null Hypothesis: H_0 : There is no significant effect on the medication

Alternate Hypothesis: H_1 : There is a significant effect on the medication

```
from scipy.stats import ttest_ind
```

```
static_value, p_value = ttest_ind(medication, placebo)
```

If $p_value \leq 0.05$, then reject H_0 , which means there is a significant effect on the medication

else then accept H_0 , which means there is no significant effect on the medication.